

# Fazle Rabbi Zaki

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## Education

### University of Michigan

Ann Arbor, MI

*B.S.E. in Aerospace Engineering*

*May 2026*

- University Honors (103rd Annual Honors Convocation); Dean's Honor List: Fall 2025
- Coursework: Attitude Determination & Control Systems (grad), Engineering for Space Environment (grad), Automatic Control, Space System Design, Spacecraft Dynamics, Propulsion, Aerodynamics

## Technical Skills

|                                |                                                                                                                                                                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GNC/ADCS:</b>               | Rigid-body dynamics, quaternion kinematics, state estimation (EKF, TRIAD), PID/LQR control, B-dot detumble, momentum management, mode logic for autonomous operations, Monte Carlo validation |
| <b>Software:</b>               | Python (NumPy, SciPy, pandas), MATLAB/Simulink, C++, Arduino IDE, Git, Linux                                                                                                                  |
| <b>Test &amp; Integration:</b> | Hardware-in-the-loop (HIL) testbed design, Software-in-the-loop (SIL) simulation, functional acceptance testing, sensor checkout, DAQ systems, oscilloscope, UART/I2C/SPI bus debugging       |
| <b>Systems Engineering:</b>    | Requirements flowdown & traceability, mass/power/cost budgeting, trade studies, interface definition, NASA Phase B PDR experience                                                             |
| <b>CAD/Analysis:</b>           | SolidWorks, CATIA, STAR-CCM+ (CFD)                                                                                                                                                            |
| <b>Hardware:</b>               | Raspberry Pi, Arduino, IMU (TDK ICM-20948), magnetometers, reaction wheels, magnetorquers, thermocouples                                                                                      |

## Engineering Projects

### CubeSat ADCS Imaging Testbed

Ann Arbor, MI

*ADCS & Systems Lead | University of Michigan (Graduate-Level Course)*

*Aug 2025 – Dec 2025*

- Developed 2-state **EKF estimator** fusing gyroscope and magnetometer data for real-time attitude determination on a 1U CubeSat prototype (0.95 kg); implemented TRIAD algorithm as initialization reference
- Built 5-state **Simulink attitude controller** (reaction wheels + magnetorquers) with autonomous **mode transitions** from detumble to nadir pointing to imaging, achieving  $<1^\circ$  RMS pointing and 50 s detumble
- Designed and executed **HIL test campaign** using Helmholtz coil testbed (168  $\mu$ T,  $\leq 10\%$  reproducibility) and rotating platform to validate estimator and controller performance in closed-loop operation
- Developed **Python ground station** (Plotly Dash, UDP) streaming telemetry at 10 Hz for real-time monitoring of estimator states, actuator commands, and sensor health during HIL test runs
- Led systems engineering for 15-member team: derived Level-1 requirements, maintained mass/power/cost budgets, and delivered \$3,000 total system cost vs. \$6,000 cap

### Icarus Heliophysics Mission

Ann Arbor, MI

*Systems Engineering & ADCS | Safety, Policy & Risk Lead*

*Jan 2026 – Apr 2026*

- Contributed to GNC **sensor trade study** evaluating sun sensors, magnetometers, star trackers, and IMUs against accuracy, mass, power, and cost metrics for a zero-momentum, three-axis stabilized architecture
- Supported **requirements flowdown** from mission-level objectives to subsystem specifications across 10 subsystems; contributed to mass/power budgeting and interface definition through **NASA Phase B PDR**
- Performed safety and risk analysis: mapped 12 rights-holders and identified 3 mission-critical risks including SEP transit exposure and launch window constraints for dual-spacecraft heliophysics mission
- Ensured regulatory compliance with NPR 8715.6 (debris mitigation), ITAR/DDTC (export control), and FCC/ITU (spectrum allocation) for deep-space X/Ka-band communications

### MASA Limelight – Hybrid Rocket Avionics

Ann Arbor, MI

*Avionics Subteam | Michigan Aeronautical Science Association*

*Aug 2025 – May 2026*

- Built **hardware-in-the-loop (HIL) test infrastructure** for sensor and pyro telemetry verification on hybrid rocket targeting 30,000 ft apogee; isolated telemetry-chain faults before hot-fire testing
- Wrote **embedded C++ firmware** for thermocouple-based thermal monitoring of propulsion components; debugged recovery valve actuation circuits using oscilloscope and DAQ
- Executed **cold-flow test operations** at MCity propulsion facility; configured ground support equipment and ran E-Box sensor checkout prior to live propellant testing

### Horizontal-Axis Wind Turbine Generator

Ann Arbor, MI

*Design Engineer | University of Michigan*

*Jan 2024 – Apr 2024*

- Designed and built self-yawing 1.54 m HAWT on \$110 budget; **verified structural integrity** through five safety inspections including 300 N lateral load test, enabling rooftop deployment approval
- Instrumented and analyzed performance (TSR up to 7.2) using DAQ and oscilloscope; diagnosed drivetrain losses and recommended 1:3 gearing; generated 6.45 V peak output during rooftop test